



Name: _____

Date: _____

Score: / 30

Learning Objectives

- Compute mean, median, mode, and range of a data set
- Compute classical (theoretical) probability
- Count outcomes using permutations and combinations
- Read and interpret frequency tables and graphs

Simplify each expression completely. Show all steps and circle your final answer.

Combinations $C(n,r)$

1. A committee of 2 people is to be chosen from a group of 8 people. Order does not matter. How many committees are possible?

$$C(8, 2) = \frac{8!}{2! 6!}$$

Answer: _____

2. A student must choose 4 questions to answer from a test that has 8 questions. How many ways can the student select the questions?

$$C(8, 4) = \frac{8!}{4! 4!}$$

Answer: _____

3. A committee of 4 people is to be chosen from a group of 10 people. Order does not matter. How many committees are possible?

$$C(10, 4) = \frac{10!}{4! 6!}$$

Answer: _____

4. A student must choose 3 questions to answer from a test that has 10 questions. How many ways can the student select the questions?

$$C(10, 3) = \frac{10!}{3! 7!}$$

Answer: _____



5. A committee of 3 people is to be chosen from a group of 11 people. Order does not matter. How many committees are possible?

$$C(11, 3) = \frac{11!}{3! 8!}$$

Answer: _____

6. A student must choose 3 questions to answer from a test that has 11 questions. How many ways can the student select the questions?

$$C(11, 3) = \frac{11!}{3! 8!}$$

Answer: _____

7. A committee of 2 people is to be chosen from a group of 7 people. Order does not matter. How many committees are possible?

$$C(7, 2) = \frac{7!}{2! 5!}$$

Answer: _____

8. A student must choose 4 questions to answer from a test that has 8 questions. How many ways can the student select the questions?

$$C(8, 4) = \frac{8!}{4! 4!}$$

Answer: _____

Mean of a data set

9. Find the mean of the data set:

{28, 38, 32, 7, 18, 29, 20, 26}

Answer: _____

10. Find the mean of the data set:

{17, 10, 19, 4, 14, 27}

Answer: _____

11. Find the mean of the data set:

{15, 14, 2, 24, 34, 27}

Answer: _____



12. Find the mean of the data set:

{16, 18, 14, 22, 9, 25, 26}

Answer: _____

Median of a data set

13. Find the median of the data set:

{8, 13, 15, 16, 18, 19, 40}

Answer: _____

14. Find the median of the data set:

{1, 6, 8, 14, 27, 37}

Answer: _____

15. Find the median of the data set:

{7, 16, 28, 32, 34, 35, 38, 39}

Answer: _____

Mode of a data set

16. Find the mode:

{3, 3, 3, 4, 14, 15, 16, 17, 20}

Answer: _____

17. Find the mode:

{1, 11, 13, 14, 15, 17, 18, 18}

Answer: _____

18. Find the mode:

{3, 3, 6, 8, 9, 10, 15, 17}

Answer: _____

Range

19. Find the range:

{41, 18, 50, 16, 48, 13, 43}

Answer: _____



20. Find the range:

{23, 21, 29, 26, 48}

Answer: _____

21. Find the range:

{14, 27, 48, 2, 10, 7, 11, 23}

Answer: _____

Permutations $P(n,r)$

22. In how many ways can 3 students be chosen from a group of 10 students and arranged in a line (order matters)?

$$P(10, 3) = \frac{10!}{7!}$$

Answer: _____

23. A club needs to elect a President, Vice-President, and Secretary from 8 candidates. No person may hold more than one office. How many different officer lineups are possible?

$$P(8, 3) = \frac{8!}{5!}$$

Answer: _____

24. In how many ways can 3 students be chosen from a group of 6 students and arranged in a line (order matters)?

$$P(6, 3) = \frac{6!}{3!}$$

Answer: _____

25. A club needs to elect a President, Vice-President, and Secretary from 10 candidates. No person may hold more than one office. How many different officer lineups are possible?

$$P(10, 3) = \frac{10!}{7!}$$

Answer: _____

26. In how many ways can 3 students be chosen from a group of 6 students and arranged in a line (order matters)?

$$P(6, 3) = \frac{6!}{3!}$$

Answer: _____



27. A club needs to elect a President, Vice-President, and Secretary from 8 candidates. No person may hold more than one office. How many different officer lineups are possible?

$$P(8, 3) = \frac{8!}{5!}$$

Answer: _____

Classical (theoretical) probability

28. A bag contains 9 red, 6 blue, and 7 green marbles. One marble is drawn at random. Find P(red).

red = 9, blue = 6, green = 7

Answer: _____

29. A bag contains 5 red, 3 blue, and 9 green marbles. One marble is drawn at random. Find P(red).

red = 5, blue = 3, green = 9

Answer: _____

30. A bag contains 9 red, 3 blue, and 5 green marbles. One marble is drawn at random. Find P(red).

red = 9, blue = 3, green = 5

Answer: _____





Topics: Classical (theoretical) probability, Mean of a data set, Mode of a data set, Median of a data set, Combinations $C(n,r)$, Permutations $P(n,r)$, Range. All answers verified by independent computation.

Solutions



Combinations $C(n,r)$

1. A committee of 2 people is to be chosen from a group of 8 people. Order does not matter. How many committees are possible?

$$C(8, 2) = \frac{8!}{2! 6!}$$

→ Order does not matter, so use combinations: $C(n,r) = n!/(r!(n-r)!)$.

→ $C(8,2) = 8!/(2! * 6!) = 28$.

Answer: $C(8, 2) = 28$

2. A student must choose 4 questions to answer from a test that has 8 questions. How many ways can the student select the questions?

$$C(8, 4) = \frac{8!}{4! 4!}$$

→ Selecting questions where order doesn't matter uses combinations.

→ $C(8,4) = 8!/(4! * 4!) = 70$.

Answer: $C(8, 4) = 70$

3. A committee of 4 people is to be chosen from a group of 10 people. Order does not matter. How many committees are possible?

$$C(10, 4) = \frac{10!}{4! 6!}$$

→ Order does not matter, so use combinations: $C(n,r) = n!/(r!(n-r)!)$.

→ $C(10,4) = 10!/(4! * 6!) = 210$.

Answer: $C(10, 4) = 210$

4. A student must choose 3 questions to answer from a test that has 10 questions. How many ways can the student select the questions?

$$C(10, 3) = \frac{10!}{3! 7!}$$

→ Selecting questions where order doesn't matter uses combinations.

→ $C(10,3) = 10!/(3! * 7!) = 120$.

Answer: $C(10, 3) = 120$

5. A committee of 3 people is to be chosen from a group of 11 people. Order does not matter. How many committees are possible?

$$C(11, 3) = \frac{11!}{3! 8!}$$

→ Order does not matter, so use combinations: $C(n,r) = n!/(r!(n-r)!)$.

→ $C(11,3) = 11!/(3! * 8!) = 165$.

Answer: $C(11, 3) = 165$



6. A student must choose 3 questions to answer from a test that has 11 questions. How many ways can the student select the questions?

$$C(11, 3) = \frac{11!}{3! 8!}$$

→ Selecting questions where order doesn't matter uses combinations.

$$\rightarrow C(11, 3) = 11!/(3! * 8!) = 165.$$

Answer: $C(11, 3) = 165$

7. A committee of 2 people is to be chosen from a group of 7 people. Order does not matter. How many committees are possible?

$$C(7, 2) = \frac{7!}{2! 5!}$$

→ Order does not matter, so use combinations: $C(n, r) = n!/(r!(n-r)!)$.

$$\rightarrow C(7, 2) = 7!/(2! * 5!) = 21.$$

Answer: $C(7, 2) = 21$

8. A student must choose 4 questions to answer from a test that has 8 questions. How many ways can the student select the questions?

$$C(8, 4) = \frac{8!}{4! 4!}$$

→ Selecting questions where order doesn't matter uses combinations.

$$\rightarrow C(8, 4) = 8!/(4! * 4!) = 70.$$

Answer: $C(8, 4) = 70$



Mean of a data set

9. Find the mean of the data set:

{28, 38, 32, 7, 18, 29, 20, 26}

→ Add all 8 values: 198.

→ Divide by $n = \{n\}$: $\{sum_values\} \div \{n\} = \{answer\}$.

Answer: $\bar{x} = 24.75$

10. Find the mean of the data set:

{17, 10, 19, 4, 14, 27}

→ Add all 6 values: 91.

→ Divide by $n = \{n\}$: $\{sum_values\} \div \{n\} = \{answer\}$.

Answer: $\bar{x} = 15.1667$

11. Find the mean of the data set:

{15, 14, 2, 24, 34, 27}

→ Add all 6 values: 116.

→ Divide by $n = \{n\}$: $\{sum_values\} \div \{n\} = \{answer\}$.

Answer: $\bar{x} = 19.3333$

12. Find the mean of the data set:

{16, 18, 14, 22, 9, 25, 26}

→ Add all 7 values: 130.

→ Divide by $n = \{n\}$: $\{sum_values\} \div \{n\} = \{answer\}$.

Answer: $\bar{x} = 18.5714$



Median of a data set

13. Find the median of the data set:

$\{8, 13, 15, 16, 18, 19, 40\}$

→ Sort the data: $[8, 13, 15, 16, 18, 19, 40]$.

→ $n = 7$ — odd — take the middle value.

→ Median = {answer}.

Answer: Median = 16

14. Find the median of the data set:

$\{1, 6, 8, 14, 27, 37\}$

→ Sort the data: $[1, 6, 8, 14, 27, 37]$.

→ $n = 6$ — even — average the two middle values (8 and 14).

→ Median = {answer}.

Answer: Median = 11

15. Find the median of the data set:

$\{7, 16, 28, 32, 34, 35, 38, 39\}$

→ Sort the data: $[7, 16, 28, 32, 34, 35, 38, 39]$.

→ $n = 8$ — even — average the two middle values (32 and 34).

→ Median = {answer}.

Answer: Median = 33



Mode of a data set

16. Find the mode:

{3, 3, 3, 4, 14, 15, 16, 17, 20}

→ Tally each value's frequency.

→ 3 appears 3 times — more than any other value.

→ Mode = {answer}.

Answer: Mode = 3

17. Find the mode:

{1, 11, 13, 14, 15, 17, 18, 18}

→ Tally each value's frequency.

→ 18 appears 2 times — more than any other value.

→ Mode = {answer}.

Answer: Mode = 18

18. Find the mode:

{3, 3, 6, 8, 9, 10, 15, 17}

→ Tally each value's frequency.

→ 3 appears 2 times — more than any other value.

→ Mode = {answer}.

Answer: Mode = 3



Range

19. Find the range:

{41, 18, 50, 16, 48, 13, 43}

→ $Range = max - min$.

→ $max = 50, min = 13$.

→ $Range = \{max_val\} - \{min_val\} = \{answer\}$.

Answer: Range = 37

20. Find the range:

{23, 21, 29, 26, 48}

→ $Range = max - min$.

→ $max = 48, min = 21$.

→ $Range = \{max_val\} - \{min_val\} = \{answer\}$.

Answer: Range = 27

21. Find the range:

{14, 27, 48, 2, 10, 7, 11, 23}

→ $Range = max - min$.

→ $max = 48, min = 2$.

→ $Range = \{max_val\} - \{min_val\} = \{answer\}$.

Answer: Range = 46



Permutations $P(n,r)$

22. In how many ways can 3 students be chosen from a group of 10 students and arranged in a line (order matters)?

$$P(10, 3) = \frac{10!}{7!}$$

→ Since order matters, use permutations: $P(n,r) = n!/(n-r)!$

→ $P(10,3) = 10!/(7)! = 720$.

Answer: $P(10, 3) = 720$

23. A club needs to elect a President, Vice-President, and Secretary from 8 candidates. No person may hold more than one office. How many different officer lineups are possible?

$$P(8, 3) = \frac{8!}{5!}$$

→ Each office is distinct, so order matters: use $P(n,r)$.

→ $P(8,3) = 8!/(5)! = 336$.

Answer: $P(8, 3) = 336$

24. In how many ways can 3 students be chosen from a group of 6 students and arranged in a line (order matters)?

$$P(6, 3) = \frac{6!}{3!}$$

→ Since order matters, use permutations: $P(n,r) = n!/(n-r)!$

→ $P(6,3) = 6!/(3)! = 120$.

Answer: $P(6, 3) = 120$

25. A club needs to elect a President, Vice-President, and Secretary from 10 candidates. No person may hold more than one office. How many different officer lineups are possible?

$$P(10, 3) = \frac{10!}{7!}$$

→ Each office is distinct, so order matters: use $P(n,r)$.

→ $P(10,3) = 10!/(7)! = 720$.

Answer: $P(10, 3) = 720$

26. In how many ways can 3 students be chosen from a group of 6 students and arranged in a line (order matters)?

$$P(6, 3) = \frac{6!}{3!}$$

→ Since order matters, use permutations: $P(n,r) = n!/(n-r)!$

→ $P(6,3) = 6!/(3)! = 120$.

Answer: $P(6, 3) = 120$



27. A club needs to elect a President, Vice-President, and Secretary from 8 candidates. No person may hold more than one office. How many different officer lineups are possible?

$$P(8, 3) = \frac{8!}{5!}$$

→ Each office is distinct, so order matters: use $P(n, r)$.

→ $P(8, 3) = 8!/(5!) = 336$.

Answer: $P(8, 3) = 336$



Classical (theoretical) probability

28. A bag contains 9 red, 6 blue, and 7 green marbles. One marble is drawn at random. Find $P(\text{red})$.

red = 9, blue = 6, green = 7

→ Count the total outcomes: $9 + 6 + 7 = 22$.

→ $P(\text{red}) = \text{favorable} / \text{total} = 9/22 = 9/22$.

→ As a decimal: $P(\text{red}) = 0.4091$ (approx).

Answer: $P(\text{red}) = \frac{9}{22} = 9/22 \approx 0.4091$

29. A bag contains 5 red, 3 blue, and 9 green marbles. One marble is drawn at random. Find $P(\text{red})$.

red = 5, blue = 3, green = 9

→ Count the total outcomes: $5 + 3 + 9 = 17$.

→ $P(\text{red}) = \text{favorable} / \text{total} = 5/17 = 5/17$.

→ As a decimal: $P(\text{red}) = 0.2941$ (approx).

Answer: $P(\text{red}) = \frac{5}{17} = 5/17 \approx 0.2941$

30. A bag contains 9 red, 3 blue, and 5 green marbles. One marble is drawn at random. Find $P(\text{red})$.

red = 9, blue = 3, green = 5

→ Count the total outcomes: $9 + 3 + 5 = 17$.

→ $P(\text{red}) = \text{favorable} / \text{total} = 9/17 = 9/17$.

→ As a decimal: $P(\text{red}) = 0.5294$ (approx).

Answer: $P(\text{red}) = \frac{9}{17} = 9/17 \approx 0.5294$

